

Introduction

We hope to solve the equation

$$i\partial_t\psi = (-i\sigma_x\partial_x + m\sigma_z + f(x))\psi$$

where $\psi = (\psi_1\psi_2)^T$, and f is some potential function. In theory, this is only slightly different from Schrödinger. If we can store the spinor ψ , nearly all the same methods can be used to solve Dirac; we will use Strang splitting to approximate the solution by solving each term individually. Note that $\propto$ can usually be interpreted as "equal up to normalization."

Splitting

Like in Schrödinger, we approximate the solution using splitting of the unitary transformation, but this time we use the Strang splitting:

$$\psi(x, t_{j+1}) = e^{-\partial_x\sigma_x\frac{\Delta t}{2}} e^{-i(m\sigma_z+f(x))\Delta t} e^{-\partial_x\sigma_x\frac{\Delta t}{2}} \psi(x, t_j) + O(\Delta t^3)$$

With $\frac{t}{\Delta t}$ timesteps, this gives a second order in Δt accumulated error $O(\frac{t}{\Delta t}\Delta t^3) = O(t\Delta t^2)$. Also recognize that $[m\sigma_z, f(x)I] = 0$, so we can split exactly $e^{-i(m\sigma_z+f(x))\Delta t} = e^{-im\sigma_z} e^{-if(x)}$. The implementation of these operators and storage of the spinor on qubits are detailed below.

Discretization and Storage

Fortunately, the problem of storing the spinor on qubits has a simple solution: we maintain a register to store position, as in Schrödinger, and add one qubit as an indicator. We discretize ψ_1 and ψ_2 into $|\psi_1\rangle$ and $|\psi_2\rangle$ as we know how

from Schrödinger, and store the state ψ as

$$\begin{aligned}
|\psi\rangle &\propto \sum_{k=0}^{N-1} (\psi_1(x_k) |0\rangle + \psi_2(x_k) |1\rangle) \otimes |k\rangle \\
&= |0\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_1(x_k) |k\rangle \right) + |1\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_2(x_k) |k\rangle \right) \\
&\propto |0\rangle \otimes |\psi_1\rangle + |1\rangle \otimes |\psi_2\rangle
\end{aligned}$$

With the indicator as the most significant qubit, this is equivalent to concatenating the column vectors of amplitudes for $|\psi_1\rangle$ and $|\psi_2\rangle$. Regarding measurement probabilities, if we measure only the register storing position, and leave the indicator qubit unmeasured, the probability of measuring $|k\rangle$ is $p(k) \propto |\psi_1(x_k, t)|^2 + |\psi_2(x_k, t)|^2$. This is justified as follows: the state $|\psi\rangle$ has density matrix

$$\begin{aligned}
|\psi\rangle \langle\psi| &\propto (|0\rangle \otimes |\psi_1\rangle + |1\rangle \otimes |\psi_2\rangle)(\langle 0| \otimes \langle\psi_1| + \langle 1| \otimes \langle\psi_2|) \\
&= |0\rangle \langle 0| \otimes |\psi_1\rangle \langle\psi_1| + |0\rangle \langle 1| \otimes |\psi_1\rangle \langle\psi_2| + |1\rangle \langle 0| \otimes |\psi_2\rangle \langle\psi_1| + |1\rangle \langle 1| \otimes |\psi_2\rangle \langle\psi_2|
\end{aligned}$$

Measurement probabilities of the position register are given by the reduced density matrix obtained by tracing out the spinor qubit. Taking the partial trace, we get

$$\begin{aligned}
\rho &\propto |\psi_1\rangle \langle\psi_1| + |\psi_2\rangle \langle\psi_2| \\
&= \sum_{l=0}^{N-1} \sum_{j=0}^{N-1} (\psi_1(x_l) \psi_1(x_j)^* + \psi_2(x_l) \psi_2(x_j)^*) |l\rangle \langle j|
\end{aligned}$$

Thus the probability of measuring $|k\rangle$ at time t is $p(k) = \text{tr}(|k\rangle \langle k| \rho) \propto |\psi_1(x_k, t)|^2 + |\psi_2(x_k, t)|^2$ by orthonormality of the computational basis.

Transport Term

We concern ourselves with the equation $i\partial_t\psi = -i\partial_x\sigma_x\psi$. Recognizing that $\sigma_x = H\sigma_zH$, where, H is the Hadamard matrix, this system is equivalent to $\partial_t H\psi = -\partial_x\sigma_z H\psi$. Letting $\varphi := (\varphi_1, \varphi_2)^T = H\psi$, since σ_z is diagonal, the system is decoupled: $\partial_t\varphi = -\partial_x\sigma_z\varphi$. Equivalently,

$$\partial_t\varphi_1 = -\partial_x\varphi_1$$

$$\partial_t\varphi_2 = \partial_x\varphi_2$$

This is a first order linear transport equation, which has a known solution for initial conditions $\varphi_j^0(x) = \varphi_j(x, 0)$: $\varphi_1(x, t) = \varphi_1^0(x-t)$ and $\varphi_2(x, t) = \varphi_2^0(x+t)$. For the sake of implementation on a quantum computer, we solve using the Fourier transform $\mathcal{F}$ to get

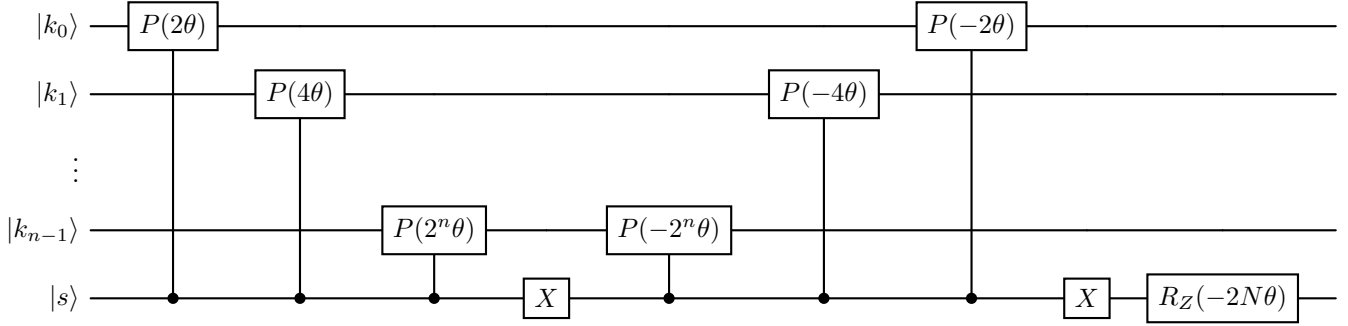
$$\varphi(x, t) = \begin{pmatrix} \mathcal{F}^{-1}(e^{-i\xi t}\mathcal{F}(\varphi_1^0)(\xi)) \\ \mathcal{F}^{-1}(e^{i\xi t}\mathcal{F}(\varphi_2^0)(\xi)) \end{pmatrix}$$

Hence on the quantum computer we map $|s\rangle \otimes |k\rangle \mapsto |s\rangle \otimes (QFT e^{(-1)^{s+1}i\xi_k\Delta t} QFT^\dagger |k\rangle)$, where as before, $\xi_k = \frac{\pi}{L}(2k - N)$. A new concern we have in Dirac is that any θ dependent global phase incurred in the application of $e^{i\theta k}$ is now a relative phase between φ_1 and φ_2 that we need to correct for. We write the phase shift operation more carefully with respect to global phase changes. Note that I've changed the implementation of $e^{ik} := \bigotimes_{j=0}^{n-1} P(2^j)$ to use phase gates, rather than differences of identity and Z gates. This makes global phase easier to keep track of.

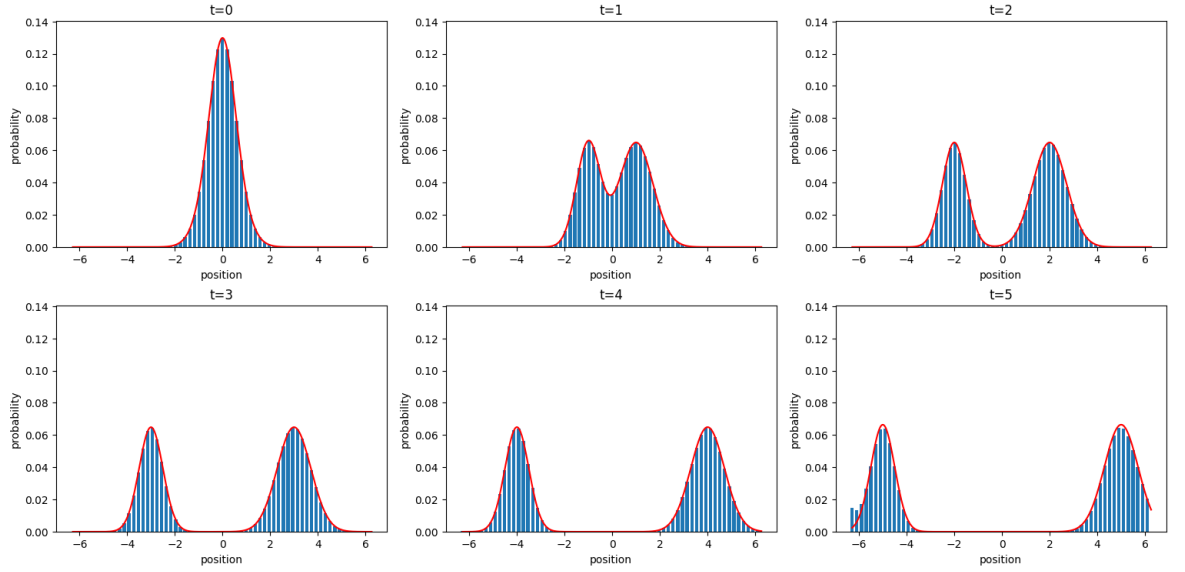
$$\exp\left((-1)^{s+1}i\frac{\pi}{L}(2k - N)\Delta t\right) = \exp\left((-1)^s i N \frac{\pi}{L}\Delta t\right) \exp\left((-1)^{s+1}2i\frac{\pi}{L}\Delta t k\right)$$

$$= \exp\left((-1)^s i N \frac{\pi}{L} \Delta t\right) \bigotimes_{j=0}^{n-1} P\left((-1)^{s+1} 2^{j+1} \frac{\pi}{L} \Delta t\right)$$

φ_1 picks up a phase of $e^{\frac{\pi i N \Delta t}{L}}$, and φ_2 picks up a phase of $e^{-\frac{\pi i N \Delta t}{L}}$, which we build in by applying $R_Z\left(\frac{-2\pi N \Delta t}{L}\right)$ to the spinor qubit. The quantum circuit we implement to apply the shift phase is below. $\theta = \frac{\pi}{L} \Delta t$.



Sandwiching this with QFT and $QFT^\dagger$ solves the linear transport equation, which is certainly interesting on its own, so I have included a plot below of a solution with two Gaussians φ_1 and φ_2 .



As interesting as this is, our concern is with Dirac, and still we need to be able to map between $|\psi\rangle$ and $|\varphi\rangle$. This is fortunately very simple and reveals itself after a bit of algebra. Recalling that $\psi = H\varphi$ and $\varphi = H\psi$, and we have the state $|\varphi\rangle \propto |0\rangle |\varphi_1\rangle + |1\rangle |\varphi_2\rangle$

$$\begin{aligned}
|\psi\rangle &\propto |0\rangle |\psi_1\rangle + |1\rangle |\psi_2\rangle \\
&= \frac{|0\rangle}{\sqrt{2}} (|\varphi_1\rangle + |\varphi_2\rangle) + \frac{|1\rangle}{\sqrt{2}} (|\varphi_1\rangle - |\varphi_2\rangle) \\
&= \frac{|0\rangle + |1\rangle}{\sqrt{2}} |\varphi_1\rangle + \frac{|0\rangle - |1\rangle}{\sqrt{2}} |\varphi_2\rangle \\
&= (H \otimes I) |\varphi\rangle
\end{aligned}$$

The same gate maps both directions, so we can transform between $|\varphi\rangle$ and $|\psi\rangle$ with a Hadamard gate on the spinor qubit. Thus our solution to $\partial_t \psi = -\sigma_x \partial_x \psi$ is given by simply applying H to the spinor qubit, following the work above for the linear transport, then applying another H . This is the circuit returned by `transport()`.

Mass Term

We now concern ourselves with the mass term $i\partial_t \psi = m\sigma_z \psi$. Fortunately, σ_z is diagonal in the standard basis, so we have the following solution.

$$\psi(x, t) = \begin{pmatrix} e^{-imt} \psi_1^0(x) \\ e^{imt} \psi_2^0(x) \end{pmatrix}$$

On qubits we want to map $|0\rangle |\psi_1\rangle + |1\rangle |\psi_2\rangle \mapsto e^{-im\Delta t} |0\rangle |\psi_1\rangle + e^{im\Delta t} |1\rangle |\psi_2\rangle$.

This is implemented with a single $R_Z(2m\Delta t)$ on the spinor qubit.

Potentials

We implement a linear potential and the quantum harmonic oscillator potential, giving a phase gate decomposition here. For $f(x) = x$, the Trotter term we implement is

$$\psi(x, t) = e^{-ixt} \psi_0(x)$$

on qubits, this looks like

$$\begin{aligned} e^{-if(x)\Delta t} : |k\rangle &\mapsto e^{-ix_k \Delta t} |k\rangle \\ &= e^{id\Delta t} e^{-ik\Delta x \Delta t} |k\rangle \\ &= \prod_{j=0}^{n-1} e^{-i2^j k_j \Delta x \Delta t} \quad \text{up to global phase} \end{aligned}$$

Hence up to global phase, this potential is given by $\bigotimes_{j=0}^{n-1} P(-2^j \Delta x \Delta t)$ on the position register. For the quantum harmonic oscillator, we have already derived a working circuit using RZ gates, but for completeness here is a derivation for the implementation in `dirac.py`, decomposing the operation with phase gates. We want to implement $e^{-ix_k^2 \Delta t} = e^{-i(-d+k\Delta x)^2 \Delta t}$, which is given by $e^{2idk\Delta x \Delta t} e^{-ik^2 \Delta x^2 \Delta t}$ up to global phase. The first term in the product is linear in k , so we know already by previous work how to decompose such a thing into phase gates. Considering the e^{-ik^2} term, we have

$$\begin{aligned} e^{-ik^2} &= e^{-i(\sum_{j=0}^{n-1} 2^j k_j)(\sum_{l=0}^{n-1} 2^l k_l)} \\ &= e^{\sum_{j=0}^{n-1} \sum_{l=0}^{n-1} 2^{j+l} k_j k_l} \\ &= e^{\sum_{j=0}^{n-1} 2^{2j} k_j} e^{\sum_{j=0}^{n-1} \sum_{l=0}^{n-1} 2^{j+l+1} k_j k_l} \end{aligned}$$

This can be decomposed into phase gates as follows.

$$e^{-ik^2} = \prod_{j=0}^{n-1} P_j(-2^{2j}) \prod_{l=j+1}^{n-1} CP_{jl}(-2^{j+l+1})$$

Finally deriving our quantum circuit for $f(x) = x^2$, we have

$$\begin{aligned} e^{-if(x_k)\Delta t} &= \prod_{j=0}^{n-1} P_j(2^{j+1}d\Delta x\Delta t)P(-2^{2j}\Delta x^2\Delta t) \prod_{l=j+1}^{n-1} CP_{jl}(-2^{j+l+1}\Delta x^2\Delta t) \\ &= \prod_{j=0}^{n-1} P_j(2^j\Delta x\Delta t(2d-2^j\Delta x)) \prod_{l=j+1}^{n-1} CP_{jl}(-2^{j+l+1}\Delta x^2\Delta t) \end{aligned}$$

Testing

Here I justify a test of the transport term. By our previous work, we know

$\psi = H\varphi$, thus the initial condition φ^0 is given by

$$\varphi^0 = H\psi^0 = \frac{1}{\sqrt{2}} \begin{pmatrix} \psi_1^0 + \psi_2^0 \\ \psi_1^0 - \psi_2^0 \end{pmatrix}$$

Solving now for ψ ,

$$\begin{aligned} \psi(x, t) &= H\varphi(x, t) \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} \varphi_1(x, t) + \varphi_2(x, t) \\ \varphi_1(x, t) - \varphi_2(x, t) \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} \varphi_1^0(x-t) + \varphi_2^0(x+t) \\ \varphi_1^0(x-t) + \varphi_2^0(x+t) \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} \psi_1^0(x-t) + \psi_2^0(x-t) + \psi_1^0(x+t) - \psi_2^0(x+t) \\ \psi_1^0(x-t) + \psi_2^0(x-t) - \psi_1^0(x+t) + \psi_2^0(x+t) \end{pmatrix} \end{aligned}$$

Thus from any initial condition ψ^0 , we can construct exact solutions according to the scheme above and discretize on qubits as $|\psi_{\text{ideal}}\rangle$. If our true $|\psi\rangle$ is successfully computed up to a global phase, we should have $|\langle\psi|\psi_{\text{ideal}}\rangle| = 1$. Defining the error $\varepsilon := |1 - |\langle\psi|\psi_{\text{ideal}}\rangle||$, we can say the test passes if ε is smaller than some predefined tolerance. Testing the mass term is much easier; since the system is already uncoupled, we can solve

$$i\partial_t\psi = m\sigma_z\psi \implies \psi(x, t) = \begin{pmatrix} e^{-imt}\psi_1^0(x) \\ e^{imt}\psi_2^0(x) \end{pmatrix}$$

and test by the same procedure.